

BIPARTITE GRAPHS ARE CONTRACTIBLE

ABSTRACT. An H -scheme assigns a path to every edge of a target graph H , with prescribed endpoints, such that paths meeting at a vertex have a common target endpoint. We prove that every scheme of a finite bipartite graph contains a minor of that graph rooted at all prescribed vertices. The proof projects a properly coloured scheme onto graphic matroids. Either disjoint spanning trees give the required model, or a minimizing set in the matroid union rank formula yields a strictly smaller root-preserving minor containing another scheme of the same target. The orientation of the bipartition may change during the induction.

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1. STATEMENT AND COLOUR NORMALIZATION

All graphs are finite. Targets and hosts are simple; auxiliary multigraphs may have parallel edges. Following Kündgen, Pelsmajer and Ramamurthi [3, Definitions 1.1 and 2.1], an H -scheme in a graph G consists of distinct roots $(r_v : v \in V(H))$ and a simple r_u, r_v -path P_{uv} for every $uv \in E(H)$, with no other root internal to a path, such that any collection of paths with a common vertex has a common target endpoint. A *rooted H -minor model* is a family of pairwise disjoint nonempty connected vertex sets $(C_v : v \in V(H))$, with $r_v \in C_v$, such that C_u, C_v are adjacent whenever $uv \in E(H)$. A graph H is *contractible* if every H -scheme contains such a model.

Theorem 1.1. *Every finite simple bipartite graph is contractible.*

The only external proof input is Edmonds' matroid union rank formula. We first give the part of the coloured-scheme reduction of [3, Lemma 3.3] needed here, without its additional degree condition. A minor is *root-preserving* when its vertices have pairwise disjoint connected preimages and each new root's preimage contains its corresponding original root.

Lemma 1.2. *Every H -scheme has a root-preserving minor whose underlying graph is properly coloured by a map f to $V(H)$, with $f(r_v) = v$, and whose scheme paths use only their endpoint colours.*

Proof. Delete unused edges and nonroot vertices. Give each root its target label. At each nonroot, choose an endpoint common to all target edges whose paths contain it. Thus each path uses only its endpoint labels. Contract every connected component induced by one label. Such a component contains at most one root; the quotient colouring is proper. Each path maps

to an endpoint-coloured walk, from which deleting closed excursions gives a simple path between the new roots. Another root cannot occur, since it has a different label. All paths meeting at a vertex have its label as a common target endpoint. Delete unused vertices and edges, retaining isolated prescribed roots. $\square$

2. A MATROID UNION EQUALITY

Let $(M_j : j \in J)$ be a finite family of multigraphs, each with distinct edge labels in a subset E_j of a finite set E . Extend its graphic matroid to E by declaring labels outside E_j to be matroid loops. Write r_j for its rank function and $M_j(X)$ for its spanning subgraph with labels in X . If $c_j(X)$ counts its components, including isolated vertices, then $r_j(X) = |V(M_j)| - c_j(X)$.

Lemma 2.1. *Let R be the maximum of $\sum_j |I_j|$ over pairwise disjoint sets $I_j \subseteq E_j$ that are forests in their respective graphs. There is $X \subseteq E$ such that*

$$(1) \quad R = |E \setminus X| + \sum_j r_j(X).$$

For any maximizing family (I_j) and any such X , the forest $I_j \cap X$ spans every component of $M_j(X)$.

Proof. Edmonds' matroid union rank formula [2, Theorem 1 and the following discussion, pp. 202–203] gives

$$R = \min_{X \subseteq E} \left(|E \setminus X| + \sum_j r_j(X) \right).$$

For a minimizing X and a maximizing disjoint family,

$$R = \sum_j |I_j \setminus X| + \sum_j |I_j \cap X| \leq |E \setminus X| + \sum_j r_j(X) = R.$$

Equality forces $|I_j \cap X| = r_j(X)$ for every j . An independent set of this rank is a spanning tree in each nontrivial component of $M_j(X)$, with isolated vertices retained. $\square$

3. A SIMULTANEOUS COMPONENT CONTRACTION

Let H be bipartite with parts (A, B) , and let G be the union of a properly endpoint-coloured H -scheme and its isolated roots, as in Lemma 1.2. Put

$$W_a = f^{-1}(a) \quad (a \in A), \quad E = f^{-1}(B) \setminus \{r_b : b \in B\}.$$

For $a \in A$, construct a multigraph M_a on W_a . Each actual occurrence of $x \in E$ on P_{ab} gives an edge labelled x between its two neighbours on that path. They are distinct vertices of colour a , so the edge is not a loop. Each label occurs at most once in M_a , since its colour determines b and the target is simple. Denote the label set of M_a by E_a .

Suppressing the internal B vertices of P_{ab} and deleting its terminal root r_b gives a path in M_a starting at r_a . These projected paths cover W_a and its edges, so M_a is connected. For an isolated target vertex it is the one-vertex graph. Every label of E is a nonloop in at least one projection.

Lemma 3.1. *Suppose $X \subseteq E$ admits pairwise disjoint forests $F_a \subseteq E_a \cap X$, each spanning every component of $M_a(X)$. Then G has a root-preserving minor Q containing an H -scheme. If $X \neq \emptyset$, then $|V(Q)| < |V(G)|$.*

Proof. For every component K of $M_a(X)$, set

$$D_{a,K} = V(K) \cup \{x \in F_a : \text{the edge labelled } x \text{ belongs to } K\}.$$

Each allocated tree lifts to two-edge paths through its actual label vertices, making $D_{a,K}$ connected. Isolated components give singleton sets. These sets are pairwise disjoint: the original A vertices belong to distinct colour components, and every allocated B label has just one owner. They contain no B root and at most one A root.

Contract each $D_{a,K}$ to $d_{a,K}$, and delete every unallocated vertex of X . Keep the roots r_b and the vertices of $E \setminus X$ as separate vertices. Retain precisely the edges from $d_{a,K}$ to surviving B vertices y witnessed by an original edge from $V(K)$ to y , deleting other edges and parallel copies. This defines a minor Q with a proper colouring: $d_{a,K}$ receives colour a , and the surviving B vertices retain their colours. The root of a is its component vertex, and every B root is unchanged.

Fix $ab \in E(H)$. Each segment uxv of P_{ab} with $x \in X$ has its two A vertices in the same component K of $M_a(X)$. Replace this traversal by a walk inside the allocated connected set $D_{a,K}$. The replacement uses only that component's allocated labels; it does not restore x through a different owner's branch set. After contraction it is just one vertex. All other steps of P_{ab} join an original A vertex to a surviving B vertex, and hence supply retained edges of Q .

Thus P_{ab} gives a walk in Q between the new roots of a, b using only colours a, b . Delete closed excursions to obtain a simple path. No other A root occurs, since it has another colour; the only B root on the original path was r_b . Every resulting path uses only its endpoint colours, so all paths meeting at a vertex have its colour as a common target endpoint. This proves the scheme condition in Q ; no scheme condition is asserted for the intermediate walks before contraction.

If X is nonempty, a label in X is a nonloop in some projection. Consequently $\sum_a r_a(X) > 0$, so an allocated forest has an edge. Its lifted two-edge path places at least three vertices in one contracted set. All other operations also contract or delete vertices, giving strictly smaller host order. $\square$

4. PROOF OF THEOREM 1.1

Remove isolated target vertices and their roots, restoring them as their original singleton branch sets afterwards. An edgeless target is immediate. For the remaining fixed target use strong induction on the host order. Lemma 1.2 gives a root-preserving minor K of the original host G , with a proper endpoint-coloured scheme and $|V(K)| \leq |V(G)|$.

Orient the bipartition (A, B) so that the number N_A of nonroots with colours in A is at most the number N_B with colours in B . Form the projections in Section 3 for K . They satisfy

$$\sum_{a \in A} r_a(E) = \sum_{a \in A} (|W_a| - 1) = N_A, \quad |E| = N_B.$$

Let (I_a) be a maximizing disjoint family of projection forests, and let $R = \sum_a |I_a|$.

If $R = N_A$, every I_a is a spanning tree. Set

$$C_a = W_a \cup I_a \quad (a \in A), \quad C_b = \{r_b\} \quad (b \in B).$$

The lifted trees make these sets connected, and the disjoint label sets make them disjoint. They retain every root. The final edge of P_{ab} joins r_b to W_a , giving every required contact.

Otherwise $R < N_A \leq N_B = |E|$. A minimizing set X in Lemma 2.1 is nonempty, since the expression at the empty set equals $|E| > R$. That lemma supplies the forests $F_a = I_a \cap X$ required by Lemma 3.1. We obtain a root-preserving minor Q containing an H -scheme, with

$$|V(Q)| < |V(K)| \leq |V(G)|.$$

The induction hypothesis supplies a rooted model in Q . Replace its vertices by their disjoint connected preimages, first in K and then in G . This preserves connectivity, disjointness, all roots and every required adjacency. The same lift applies to the direct spanning-tree case. Every recursive application strictly decreases host order, completing the induction. $\square$

The orientation can reverse at later induction steps. Accordingly, lifting the component contractions may expand roots on both original shores; no singleton-shore normal form is needed.

5. THE BIPARTITE-FLOW ASSERTION

Corollary 5.1. *Let H be bipartite of minimum degree at least two. Inject its vertices into a finite graph G as terminals and choose a simple path between the corresponding terminal pair for every target edge. If paths for edges with four distinct endpoints are vertex-disjoint, then G has an H -minor rooted at every terminal.*

Proof. A pairwise intersecting family of edges in a bipartite graph has a common endpoint: after two edges ab, ac , an edge meeting both but avoiding a would be bc , forming a triangle. Hence paths meeting at a vertex satisfy the common-endpoint condition. No terminal r_v can be internal to a path

for a nonincident edge ab : at most one of a, b is adjacent to v , so v has a neighbour w outside $\{a, b\}$. Then P_{vw} and P_{ab} meet at r_v although their edges have four distinct endpoints. Thus the paths form a scheme, and Theorem 1.1 applies. $\square$

The theorem includes every $K_{m,n}$ with all roots prescribed and gives a positive contractibility answer to the bipartite question in [3, Section 8, Question 4]. Corollary 5.1 independently proves the intended existence assertion of Biswal, Lee and Rao [1, Lemma 3.2], with every terminal retained. Here the convention is that paths for independent demand edges do not intersect, as in their preprint v2; the published definition appears to reverse the intersection wording.

Their prefix construction is not used. For example, in the coloured $K_{2,2}$ -scheme

$$P_{ij} = a_i b'_j a'_i b_j \quad (i, j \in \{0, 1\}),$$

all eight displayed vertex names are distinct. Their left prefixes are $\{a_i\}$, and their proposed right sets are $\{b_j, b'_j, a'_0, a'_1\}$, which overlap. This refutes the construction's Lemma 3.6, not its intended minor-existence assertion. Our result supplies an independent existence proof; it makes no publication-priority claim or new assertion about downstream spectral estimates. No implication to Hadwiger's conjecture is established here.

REFERENCES

- [1] P. Biswal, J. R. Lee and S. Rao, *Eigenvalue bounds, spectral partitioning, and metrical deformations via flows*, J. ACM **57**(3) (2010), Article 13. doi:10.1145/1706591.1706593. Preprint convention: arXiv:0808.0148v2.
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